HACKHARVARD 2026 · REGENERON: REINVENT CLINICAL TRIALS · CHALK

TrialLens

Every exclusion deserves evidence.

An evidence-grounded, challengeable reviewer for clinical-trial eligibility decisions: built on real protocols, verified sources and a measured evaluation.

Research prototype · not clinical or regulatory advice

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THE BOTTLENECK

Eligibility criteria are written from precedent, and reviewed too late

- Protocol teams "rarely get evidence-based feedback on their draft until it's too late to change cheaply" (Regeneron, Trial by Design Pt 2).
- **57%** of protocols have ≥1 substantial amendment; **45%** of those judged avoidable (Getz et al. 2016).
- FDA finalized **three oncology guidances in July 2026** on exactly the criteria most often templated: lab values, performance status, washout periods.

> *"Unnecessarily restrictive eligibility criteria may slow subject accrual, limit patients' access to clinical trials, and lead to trial results that do not fully represent treatment effects…"*

FDA, Cancer Clinical Trial Eligibility Criteria: Laboratory Values (July 2026), §I, lines 37–39

WHY IT'S HARD

"Loosen the criteria" misses most of the problem

Restrictions have reasons

- Patient safety (e.g. myelosuppressive chemotherapy)
- Predictive and prognostic enrichment
- Reduced heterogeneity → statistical power
- Interpretability of adverse events

FDA Enrichment Strategies (2019); Regeneron mentor insights §1.4–1.5

The evidence is scattered

EMPOWER-Lung 3 (Regeneron, NCT03409614): the registry text **never mentions** the neutrophil threshold. The lab criteria are on **page 67 of a 170-page PDF**. Comparable trials' thresholds sit in other sponsors' PDFs.

Checked live by a Chalk feature resolver: `registry_mentions_anc = false`

WHO AND WHERE

A reviewer for the protocol team, before the protocol is final

Users: clinical scientists, protocol authors, medical reviewers, cross-functional protocol-review teams. TrialLens prepares the evidence; **the team decides**.

Draft protocol
NCT ID or posted protocol PDF

→

TrialLens review
criteria · regimen · comparators · FDA · verifier

→

Review packet
evidence for and against, open questions, next action

→

Protocol review meeting
accept / reject / needs medical review / defer

→

Decision record
rationale documented in the protocol

Output is a defensible *question with evidence*, not an automated answer.

WHAT IT DOES

From protocol PDF to criterion-level evidence

Read

Extracts lab, ECOG and washout criteria with the **verbatim line and page**. Handles tables, Symbol-font glyphs, image-only pages.

Compare

Finds comparable trials (same disease, setting, histology, drug classes, **started earlier**), parses their PDFs, checks **regimen coverage**.

Ground

32 FDA passages (2026 eligibility guidances + 2019 enrichment), **verified verbatim** with section, page and line.

Classify

Review recommended
Evidence supports retaining
Insufficient evidence plus an inferred purpose and missing evidence.

Verify

Factual checks + clinical-logic rules. Outcomes: Supported · Qualified · Rejected · **Needs medical review**.

Decide

Exportable review packet with decision fields (Markdown / PDF / JSON).

Deterministic, auditable engine; optional model may only downgrade and is always verified

REAL EXAMPLE · EMPOWER-LUNG 3

Hemoglobin ≥10 g/dL: mixed evidence, so review it, don't remove it

SIDE-BY-SIDE WITH COMPARABLE TRIALS

TRIAL	REGIMEN	THRESHOLD	VS TARGET	REGIMEN COVERAGE	SOURCE
NCT03409614 (reviewed)	**cemiplimab, ipilimumab, pemetrexed, carboplatin, cisplatin, paclitaxel**	**≥10 g/dL**	—	—	p.67
CheckMate 9LA NCT03215706 · Bristol-Myers Squibb · start 2017-08-24	carboplatin, cisplatin, ipilimumab, nivolumab, paclitaxel, pemetrexed	≥9 g/dL	comparator looser	full	p.42
eNERGY NCT03351361 · Rennes University Hospital · start 2018-02-19	carboplatin, ipilimumab, nivolumab, paclitaxel, pemetrexed	>10 g/dL	same	full	p.8
CheckMate 227 NCT02477826 · Bristol-Myers Squibb · start 2015-08-05	carboplatin, cisplatin, gemcitabine, ipilimumab, nivolumab, paclitaxel, pemetrexed	≥9 g/dL	comparator looser	full	p.73
NCT02220894 NCT02220894 · Merck Sharp & Dohme LLC · start 2014-10-30	carboplatin, paclitaxel, pembrolizumab, pemetrexed	≥9 g/dL	comparator looser	full	p.33

Coverage = whether the comparator's regimen contains the same toxicity-relevant drug classes as this trial (e.g. myelosuppressive classes for hematologic criteria). Partial coverage weakens the comparison.

- **3** comparable PD-1 + platinum trials used **≥9 g/dL** (CheckMate 9LA, CheckMate 227, KEYNOTE-042)
- **eNERGY** used the same strict **>10 g/dL**
- Protocol p.51: grade 3–4 anemia higher with pemetrexed (a safety rationale)
- Toxicity justifies *a* threshold, not *this* number → **Review recommended**
- Missing: no stated reason for 10 vs 9; no outcome data by threshold

All values link to protocol pages; comparators chosen automatically

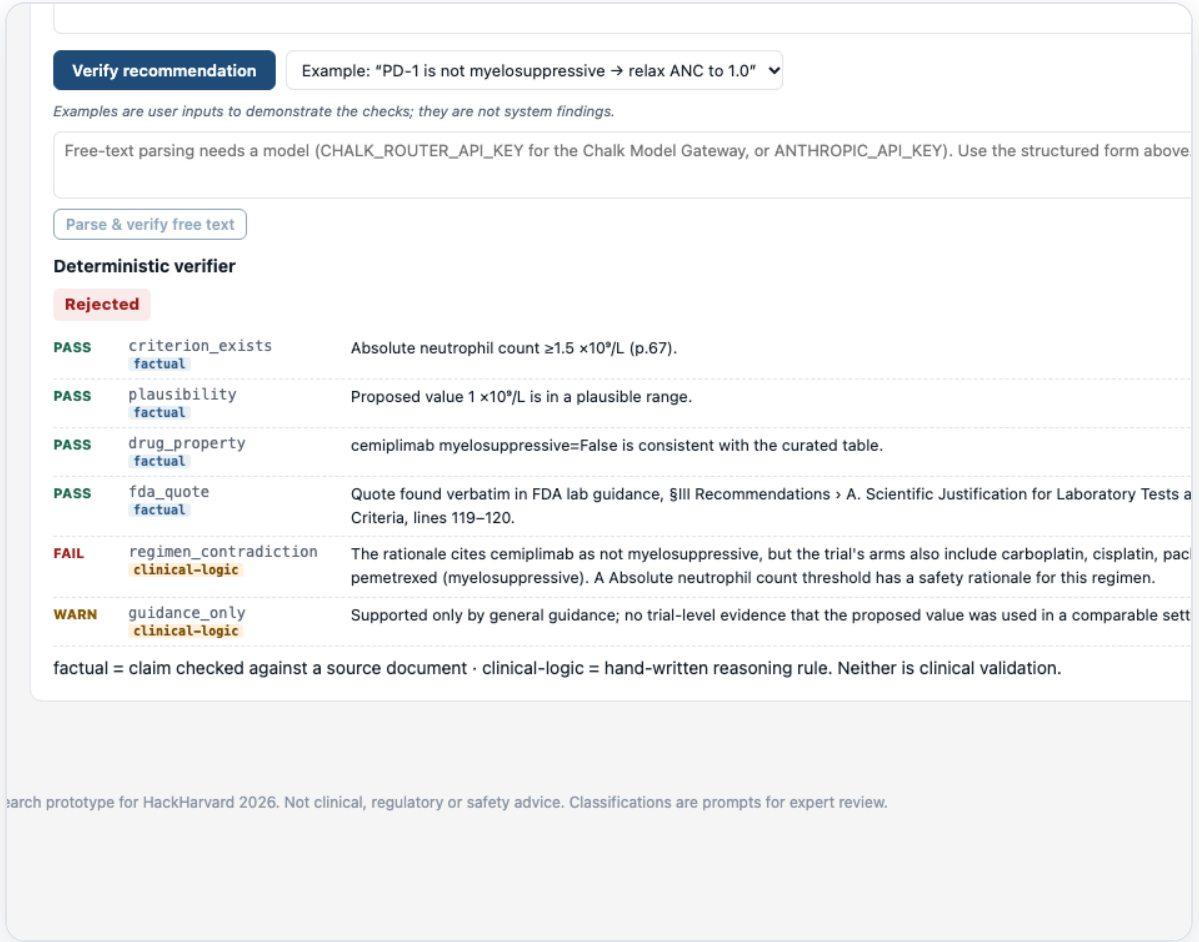
CHALLENGEABLE

The verifier rejects a plausible argument with a true premise

Challenge (typed example): "Cemiplimab is a PD-1 inhibitor and not myelosuppressive, so lower the ANC floor to 1.0."

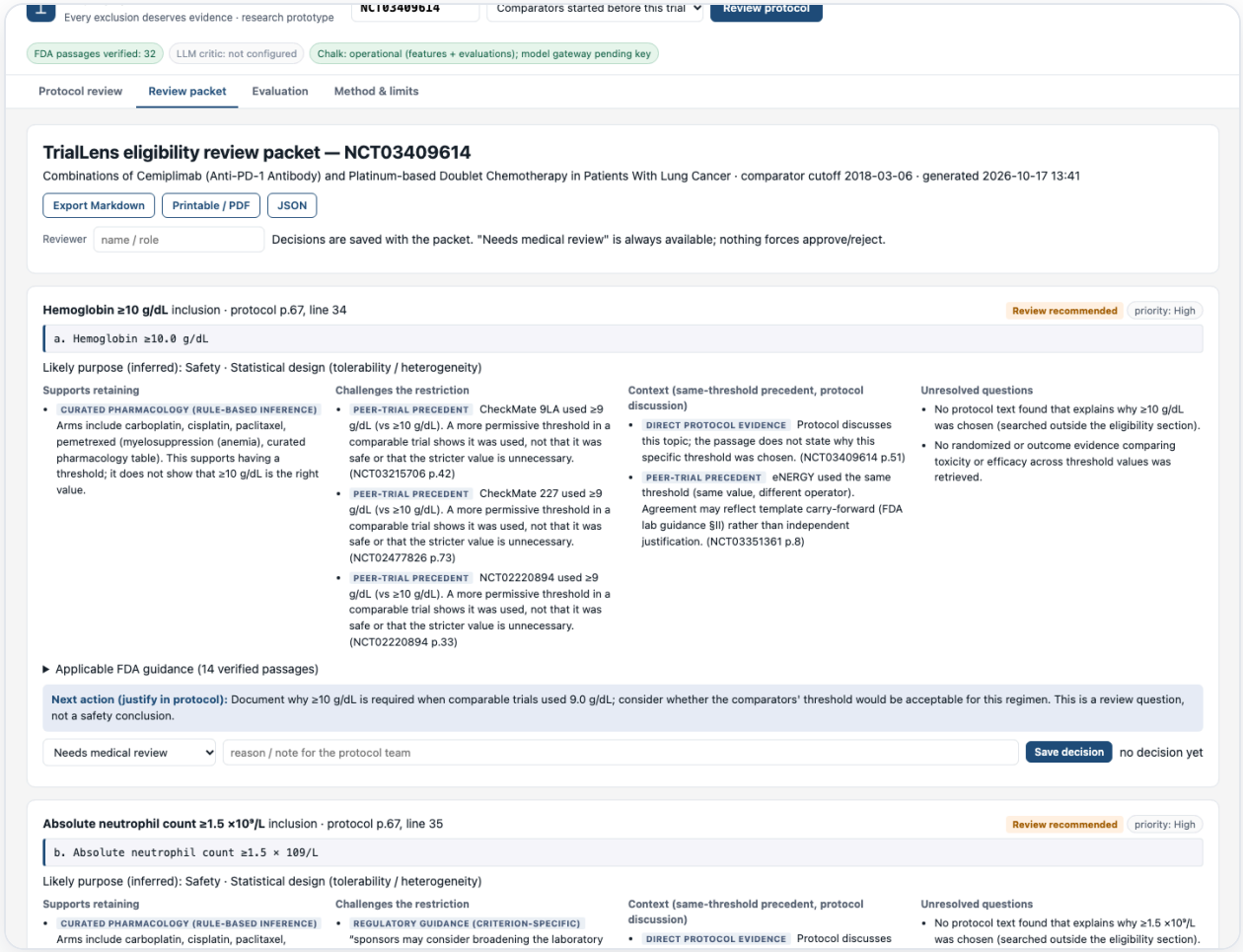
- Premise checks pass (factual)
- **regimen_contradiction** fails (clinical-logic): other arms contain carboplatin, cisplatin, paclitaxel, pemetrexed → **Rejected**
- A regimen-matched relaxation (Hgb 9.0) is not approved either: it becomes **Needs medical review**

We label which checks are factual and which are hand-written rules. Neither is clinical validation.



THE DELIVERABLE

A review packet the protocol team can act on



- Original requirement, verbatim, with page
- Priority and the reason for it
- Likely purpose: *inferred* unless sourced
- Supporting · challenging · context evidence, by tier
- Unresolved questions
- Next action
- Decision: accept · reject · **needs medical review** · defer

Export: Markdown · printable PDF · JSON

EVALUATION (NOT CLINICAL VALIDATION)

Measured on real protocols, including the ones we hadn't seen

WHAT	RESULT	NOTE
Extraction, 19 real protocols	**122/128** criteria · 97.6% precision	Unseen set 1 first run **15/21** (→21/21 after fix); unseen set 2 **29/33**, single run
Planted stricter thresholds (24, Regeneron method)	Never "supports retaining" **24/24** · escalated **16–22/24**	Depends on precedent cutoff; first run found 3 bugs, documented
Loosened thresholds falsely flagged	**0/10**	Synthetic edits of authentic protocols
FDA quotes / protocol source lines	**32/32** · **157/157**	Verbatim, page-anchored
Verifier, developer-designed faults	**24/24** rejected · **0/10** valid rejected	Same author wrote faults and verifier. Also run on Chalk.

Single annotator; no clinician review yet. Scripts in evals/ reproduce every number.

BUILT ON CHALK

Chalk does real work in TrialLens

Features + resolver

A `Trial` feature set resolved live from ClinicalTrials.gov inside Chalk. The dashboard queries it per trial: is the threshold even in the registry? **Deployed; online queries succeed.**

Compute + Evaluations

The verifier runs as a Chalk function over 39 cases, graded by Chalk scorers. **Run SUCCEEDED, 0 task errors**; 24/24 faults rejected remotely.

Model Gateway

Router-backed LLM critic (may only downgrade) and free-text challenge parsing, always gated by the verifier. **Implemented; awaiting event key.**

Evaluation `cmvco3m0n01dw01y520oc80sq` · environment `trllf247b678`

HONEST SCOPE AND WHAT'S NEXT

What TrialLens is, and isn't

Is

- A working, reproducible reviewer for lab, ECOG and washout criteria in oncology
- Complementary to criteria-drafting tools (e.g. Regeneron/Keiji ISPOR 2026): they draft from precedent; we challenge individual restrictions
- MIT-licensed, runs from a clean clone, 13 tests

Isn't (yet)

- Clinically validated: rule-based, single-annotator gold, no clinician review
- A recruitment, cost or safety predictor; we make no such claims
- **Next:** internal precedent and amendments · learning from team decisions · ICH M11 / USDM structured protocols · clinician review

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